WEEK 1 · LECTURE

Computational thinking & text as data

Dr. Philipp K. Masur
Computational Analysis of Digital Communication · Vrije Universiteit Amsterdam

A lot of data comes as text



1 / 4

A lot of data comes as text

to date on marketing opportunities!

Muddled and far too long.

Superb android app in 2021 guys ver
helpful app

ong.

REVIEW
Wow... Loved this place.

SPEECH · 1946
the Nazi-Fascist reign of terror in Europe

REVIEW
Crust is not good.

ull but never a

REVIEW
Brilliant writing and a superb cast.

TWEET
862 patients test positive for #COVID19 in Tamil Nadu today. Here is the update.

TWEET
Yesterday four people died and 216 were infected all over Japan

enty minutes.

REVIEW
Slow, but worth it.

TWEET
Mind-boggling. #Corona is a criminal conspiracy, just follow the patents.

SPEECH · 2015
years that dawned with terror touching our shores

ong.

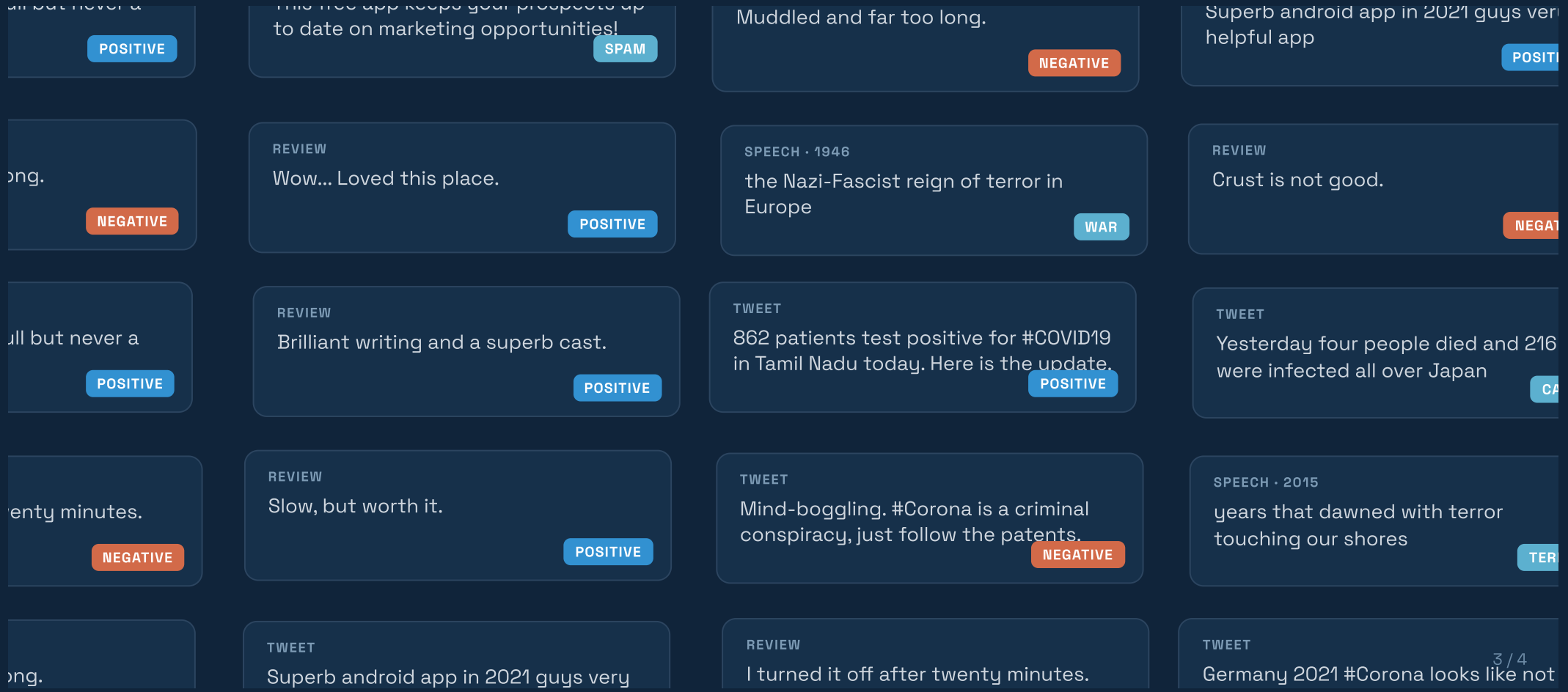
TWEET
Superb android app in 2021 guys very

REVIEW
I turned it off after twenty minutes.

TWEET
Germany 2021 #Corona looks like not

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A lot of data comes as text



A lot of data comes as text

POSITIVE

Europe

WAR

erb cast.

POSITIVE

TWEET

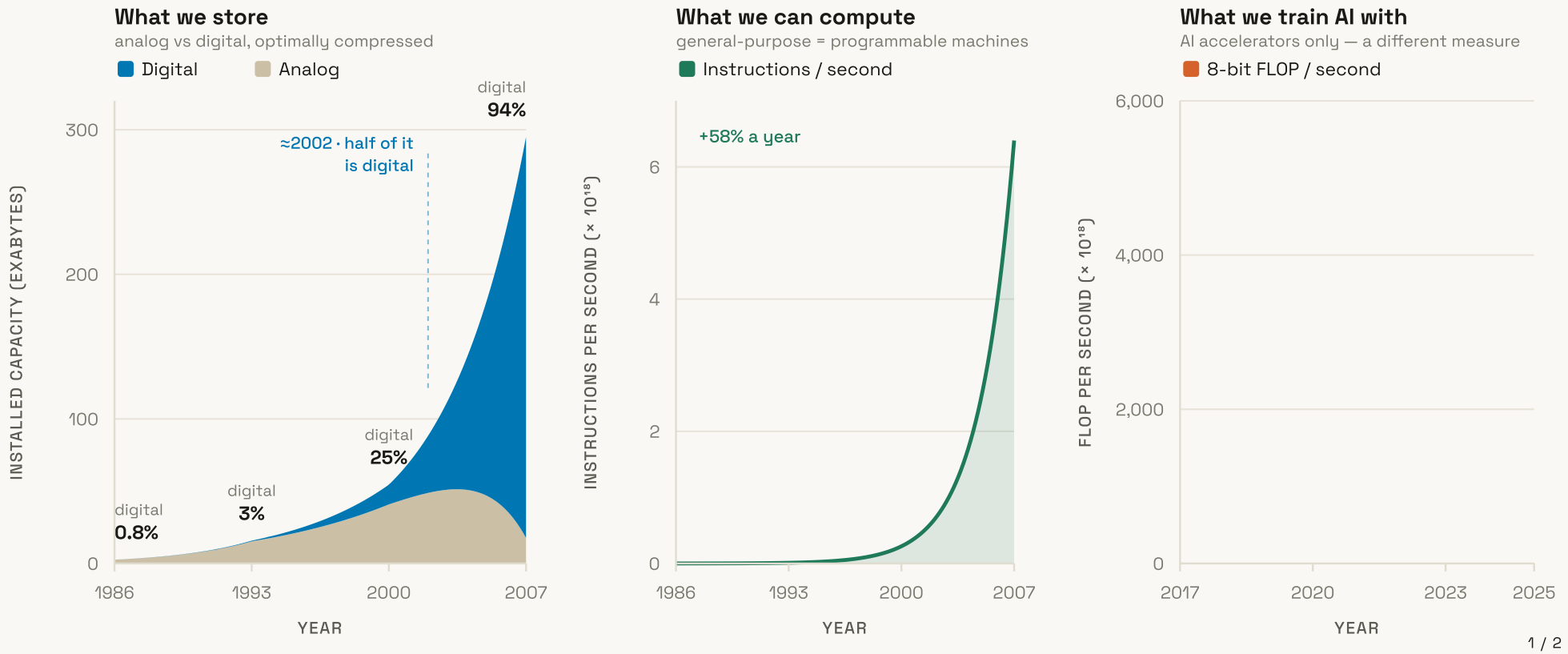
862 patients test positive for #COVID19 in Tamil Nadu today. Here is...

POSITIVE

TWEET

4 / 4

The world went digital in about twenty years



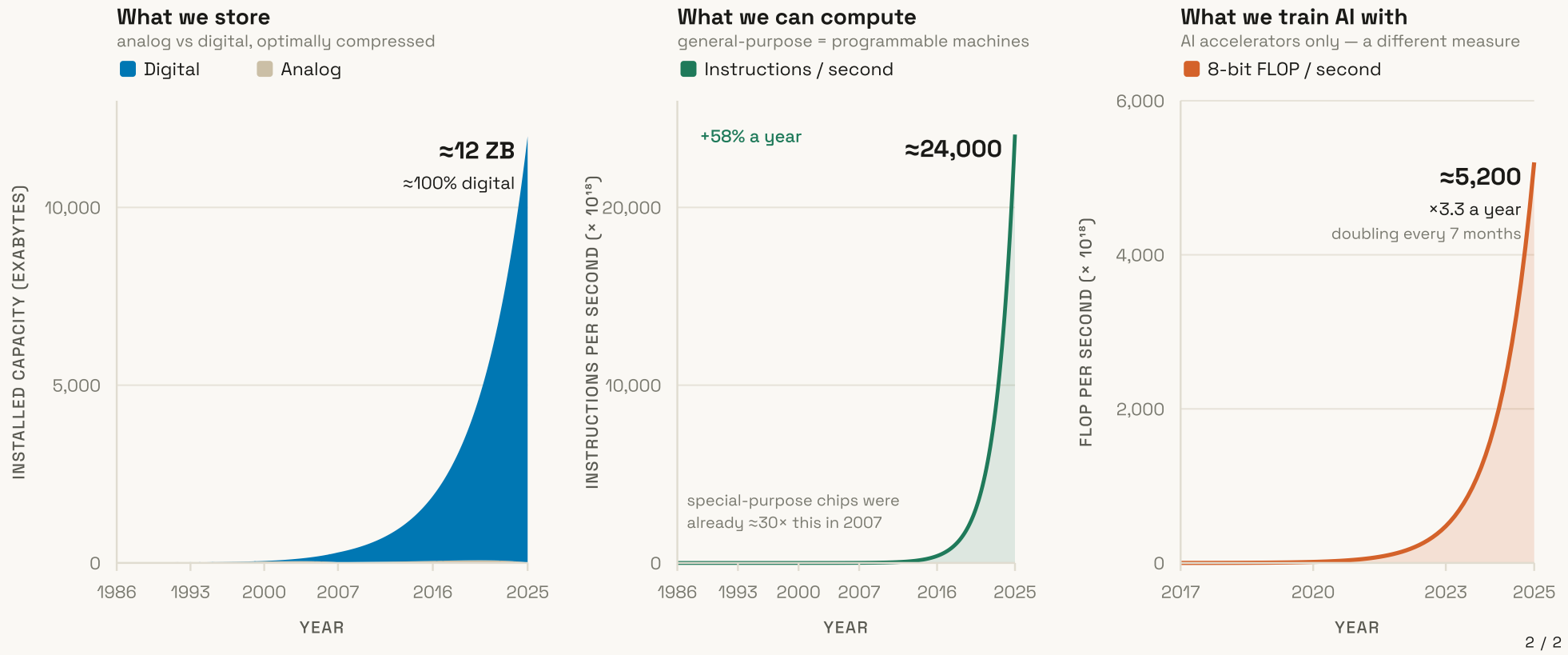
1 / 2

Hilbert, M., & López, P. (2011). The world's technological capacity to store, communicate, and compute information. *Science, 332*(6025), 60–65. https://doi.org/10.1126/science.1200970

International Data Corporation. (2024). *Global DataSphere and StorageSphere forecasts.* IDC. — post-2007 storage.

Epoch AI. (2026). *Global AI computing capacity is doubling every 7 months.* https://epoch.ai/data-insights/ai-chip-production — AI panel.

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Much of what we know about human behavior

... IS BASED ON WHAT PEOPLE TELL US

- Self-report measures in surveys
- Responses in experimental research
- Accounts in qualitative interviews

Useful for measuring motivations, attitudes, or feelings, but such measures can be biased: reported and actual behaviour diverge systematically.

... NOT BASED ON RECORDS OF ACTUAL BEHAVIOR

messages · posts · comments · reviews · news articles · wiki entries · transcripts · parliamentary debates · code

Communication that was never explicitly elicited for research, but is stored and available at scale. How can we use it?

Parry, D. A., Davidson, B. I., Sewall, C. J. R., Fisher, J. T., Mieczkowski, H., & Quintana, D. S. (2021). A systematic review and meta-analysis of discrepancies between logged and self-reported digital media use. *Nature Human Behaviour, 5*(11), 1535–1547. https://doi.org/10.1038/s41562-021-01117-5

Scharkow, M. (2016). The accuracy of self-reported internet use — A validation study using client log data. *Communication Methods and Measures, 10*(1), 13–27. https://doi.org/10.1080/19312458.2015.1118446

An standard question, but too many articles to code

- **How did Dutch news cover the housing crisis between 2015 and 2025?**
  - Did the coverage get more negative over the decade?
  - Did the framing shift from economics to fairness?
  - Whose voices were quoted: politicians, developers, tenants?
- Nothing computational about the question itself: this is the content analysis communication research has been doing since the 1950s
- To answer it you would read every article mentioning *woningmarkt* in four national outlets, 2015 to 2025

Articles to code	100,000
Minutes to read and code one	× 4
Hours	≈ 6,700
Working days	≈ 830

3 years

one coder, full time, no holidays, reading each article exactly once

The two big questions this course will answer

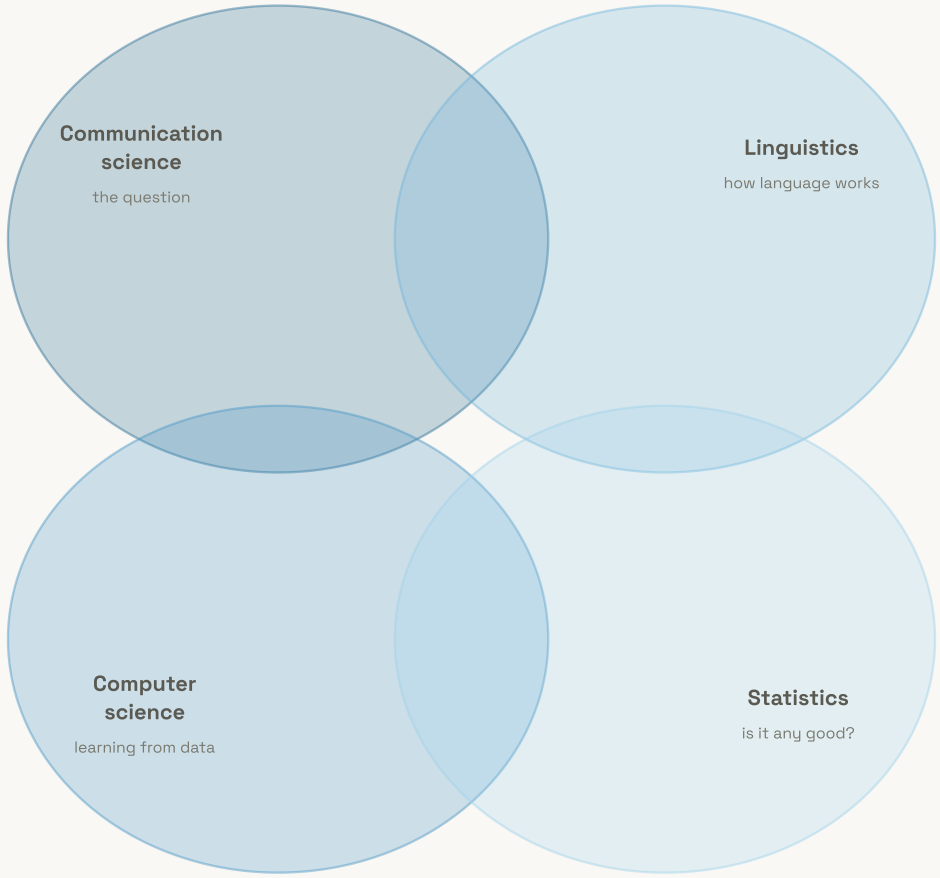
1. How can we analyse all of this text — automatically?
2. How can we make sure that the analysis is valid?

Objectives and learning goals

After completion of the course, you will...

1. **identify the data-analytic problem** in a communication question, state it precisely enough for a machine, and judge whether a computational method can answer it
2. **describe the text-classification pipeline** end to end and name where each decision can go wrong
3. **explain what each method costs and assumes**, from word counts to large language models, and when the cheap one is right
4. **treat measurement as a research question** and decide when a number is not good enough to publish
5. **run a small text-analysis study** in a group, from research question to presentation

Only the third is about specific methods. The others will outlast whatever tool is "current" when you graduate.



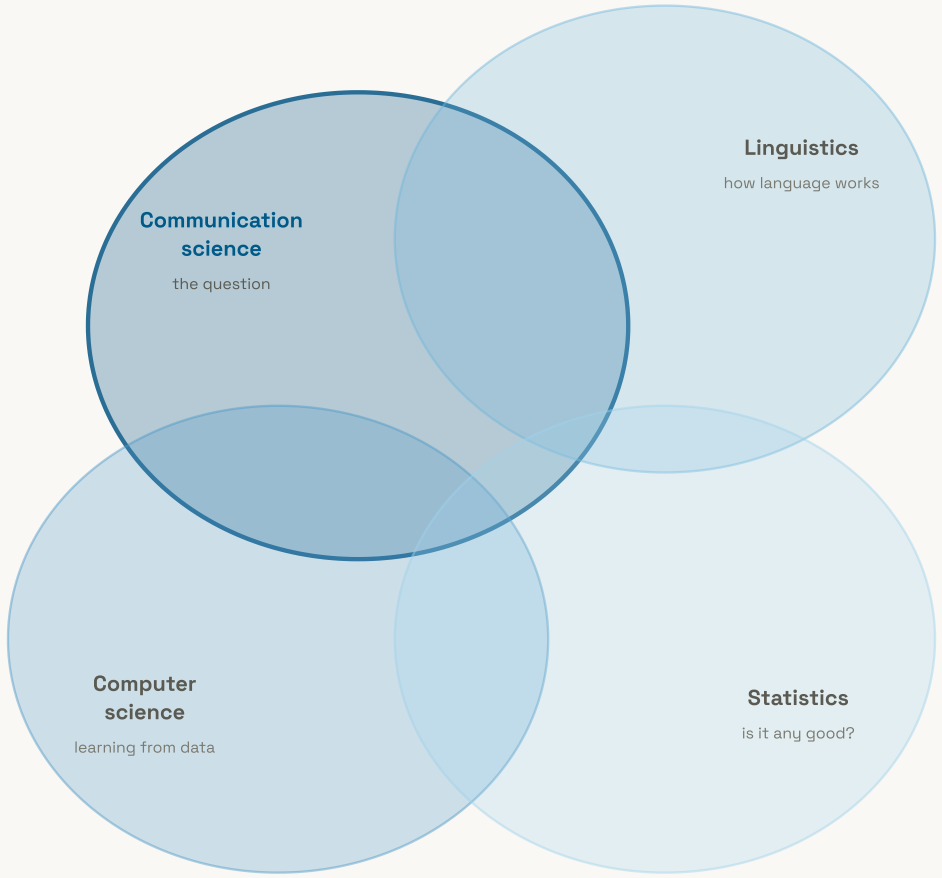
1 / 6

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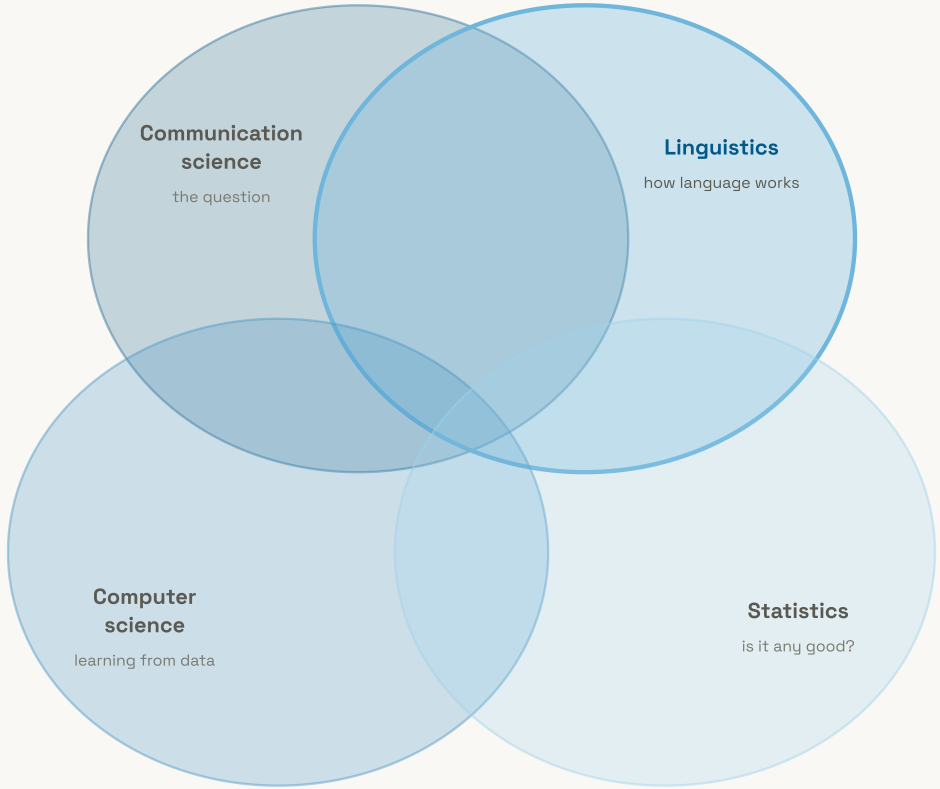
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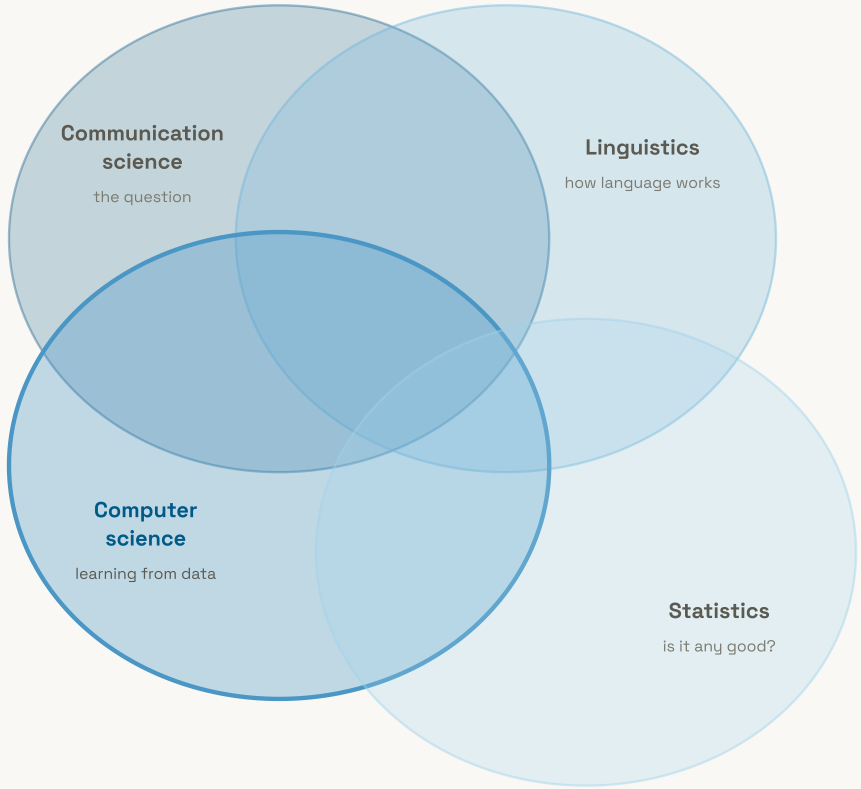
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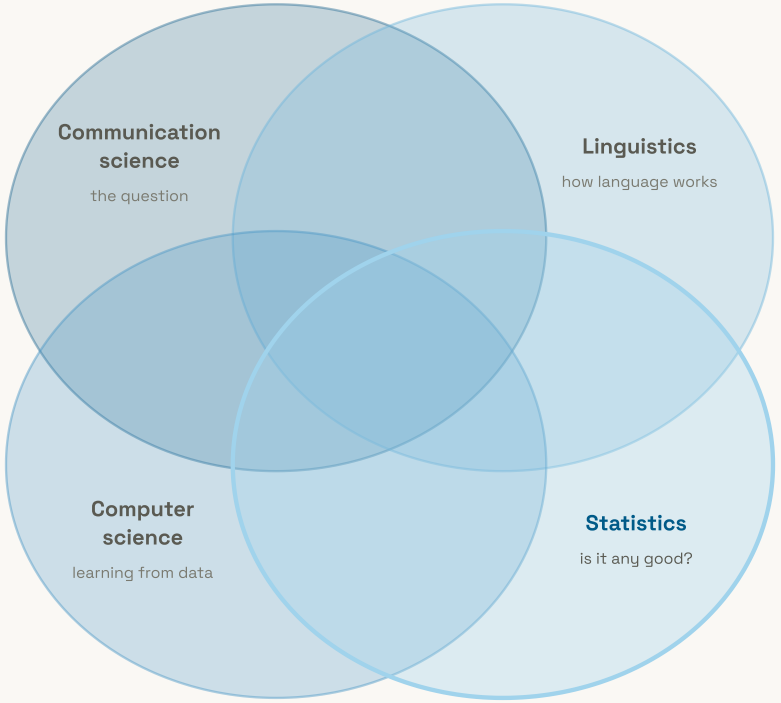


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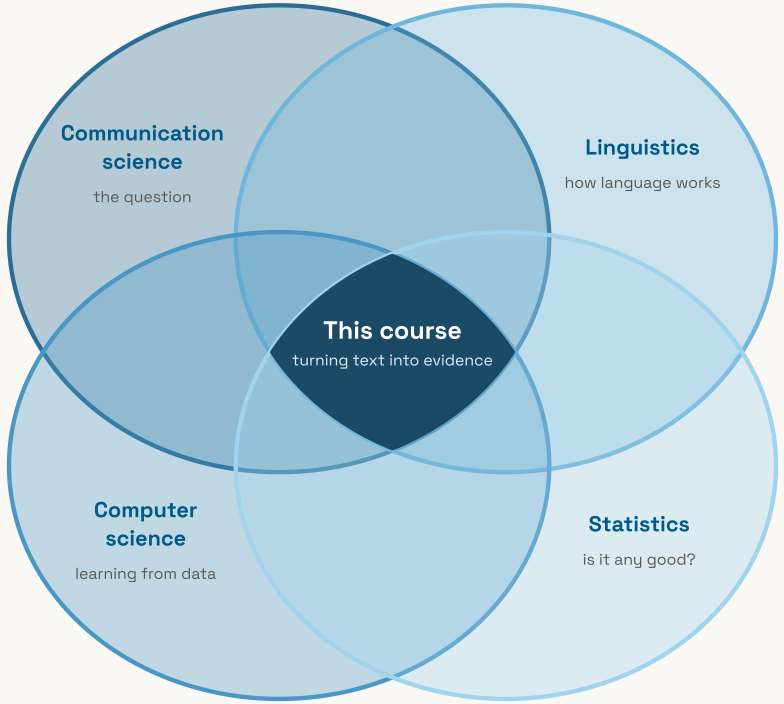


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Skills and methods

With regard to the specific methods taught in R, you will be able to…

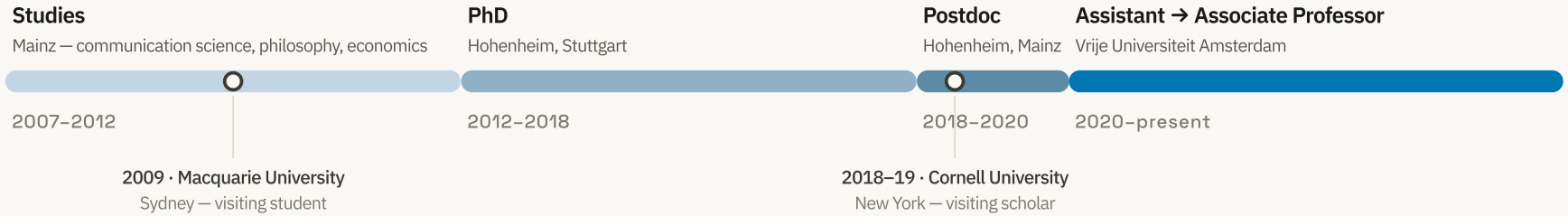
- **import and wrangle text** into a tidy table, and turn it into tokens, counts and embeddings
- **apply dictionaries, supervised classifiers, word embeddings and LLMs** to the same texts
- **build a gold standard**, compute coder agreement (Cohen's κ, Krippendorff's α), read a confusion matrix, and report precision, recall and F1
- **obtain data** for a project of your own from APIs, archives and existing corpora
- **make an analysis reproducible** from a clean start, document its decisions, and disclose what an AI tool did

> **Each of these is practised in the labs, the homework or the group project.**

Who am I?



- **Research focus** — privacy and self-disclosure online, social influence and behavioural contagion, digital literacy, social media and well-being
- **Methods** — simulation and experimental designs, scale development and validation, Bayesian statistics, computational methods and machine learning
- More at philippmasur.de



Your lab teachers



- **Dr. Noon Abdulqadir**: Research Associate, Communication Science
- **Labs**: two of the six groups, in English



- **Gianni Quaedvlieg, MSc**: Lecturer, Communication Science
- **Labs**: two of the six groups, in Dutch

> I teach the other two groups, in Dutch or English (depending on the groups' composition of students). Your group, and with it your teacher and the lab language, is assigned on Canvas.

This lecture

I

Computational communication science

What it is, why now, and what it looks like in published research

II

Text as data

What a text is to a computer, and how it becomes numbers

III

The text classification pipeline

Four steps this course keeps returning to, and how to state a problem a machine could attempt

IV

A short history

From counting words to language models — one generation per week

V

Ethics

What changes when the data were never given to you

VI

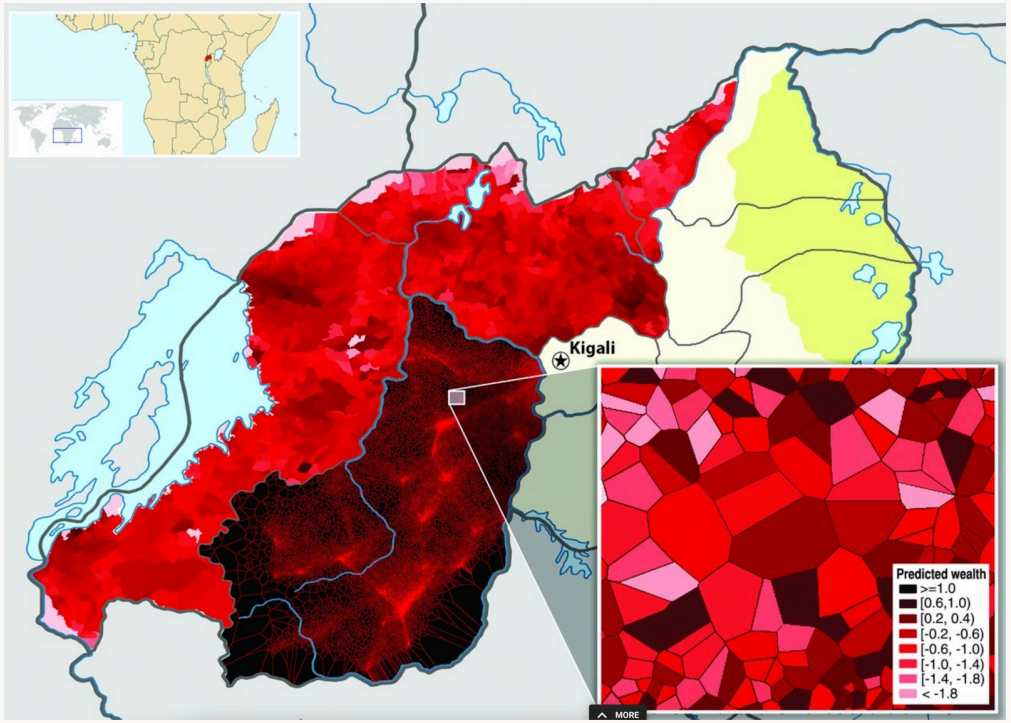
How this course works

Rhythm, assessment, and what happens this week

I · Computational communication science

An example: a surprising source of information

- 2009, Rwanda: a survey of 1,000 customers of the largest mobile provider (demographics, assets, wealth)
- The researchers also had the call records of 1.5 million subscribers
- A model trained on the 1,000 predicts wealth from calling behaviour for everyone else
- Location comes free: the towers that carried the calls



Predicted wealth by district, and by cell tower in the inset.
Blumenstock, Cadamuro & On (2015), *Science*.

Blumenstock, J., Cadamuro, G., & On, R. (2015). Predicting poverty and wealth from mobile phone metadata. *Science, 350*(6264), 1073–1076. https://doi.org/10.1126/science.aac4420

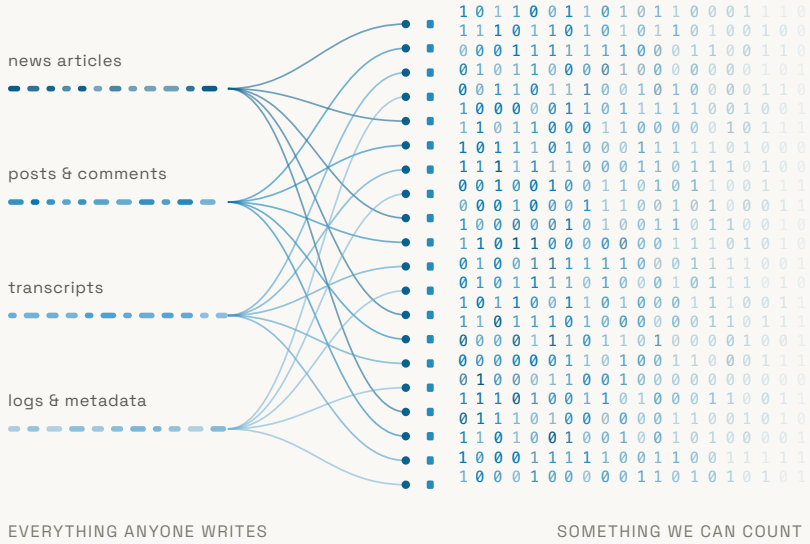
Computational social science

- Social science that uses **algorithmic tools** and large, unstructured data to study human and social behaviour
- Complements surveys, experiments and interviews rather than replacing them; the question comes first
- The usual ingredients
  - Digital traces: posts, logs, transactions, call records
  - Automated text, image, or video analysis
  - Machine learning for classification and prediction
  - Simulation of social processes

Lazer, D., Pentland, A., Adamic, L., Aral, S., Barabási, A.-L., Brewer, D., … Van Alstyne, M. (2009). Computational social science. *Science, 323*(5915), 721–723. https://doi.org/10.1126/science.1167742

Why now?

- **Big data**: social media, digital archives, digitised newspapers, logs of almost everything
- **Constant records**: traces of people and organisations accumulate as a by-product of ordinary life
- **More compute**: powerful, cheap, and available without owning a computer
- **Sophisticated tools**: text classification, topic models, word embeddings, and now large language models, much of it open source



Big data · three things that help

CHARACTERISTIC	IN ONE LINE
1 · Big	Rare events, heterogeneity, small differences become studyable
2 · Always on	Continuous collection: unexpected events, real time
3 · Non-reactive	Being measured does not change the behaviour

> **None of the three guarantees that the data are good.**

Salganik, M. J. (2018). *Bit by bit: Social research in the digital age*. Princeton University Press. Chapter 2.3.

Big data · seven things that get in the way

CHARACTERISTIC	IN ONE LINE
4 · Incomplete	Made for another purpose; your variable is missing
5 · Inaccessible	Held by companies and governments
6 · Non-representative	Users of a platform are not a population
7 · Drifting	The system changes under you
8 · Algorithmically confounded	Behaviour is shaped by the platform’s engineering
9 · Dirty	Spam, bots, duplicates, junk
10 · Sensitive	Some of it should never have been collected

Salganik, M. J. (2018). *Bit by bit: Social research in the digital age*. Princeton University Press. Chapter 2.3.

Pros and cons of computational methods

OPPORTUNITIES

- Actual behaviour instead of reports of it
- People in their own context, not a lab
- Very large N, and therefore statistical power
- Patterns that were not observable before

PITFALLS

- The techniques are demanding
- The data are often proprietary and cannot be shared
- Samples are biased in ways that are hard to see
- Little metadata; unknown provenance
- Models we no longer understand

Computational communication science

> "The emerging subfield that investigates the use of computational algorithms to gather and analyze big and often semi- or unstructured data sets to develop and test communication science theories."
>
> van Atteveldt & Peng (2018)

- Large and complex data sets
- Consisting of digital traces and other naturally occurring data
- Requiring algorithmic solutions to analyse
- Used to **develop and test communication theory**

Van Atteveldt, W., & Peng, T.-Q. (2018). When communication meets computation: Opportunities, challenges, and pitfalls in computational communication science. *Communication Methods and Measures, 12*(2–3), 81–92. https://doi.org/10.1080/19312458.2018.1458084

Where it shows up

FIELDS OF COMMUNICATION RESEARCH THAT USE COMPUTATIONAL METHODS

Political communication | Journalism | Misinformation | Platforms and algorithms

Health communication | Science communication | Climate and environment | Advertising

Strategic communication | Organisational communication | Interpersonal communication

Mobile communication | Media psychology | Entertainment and games | Children and media

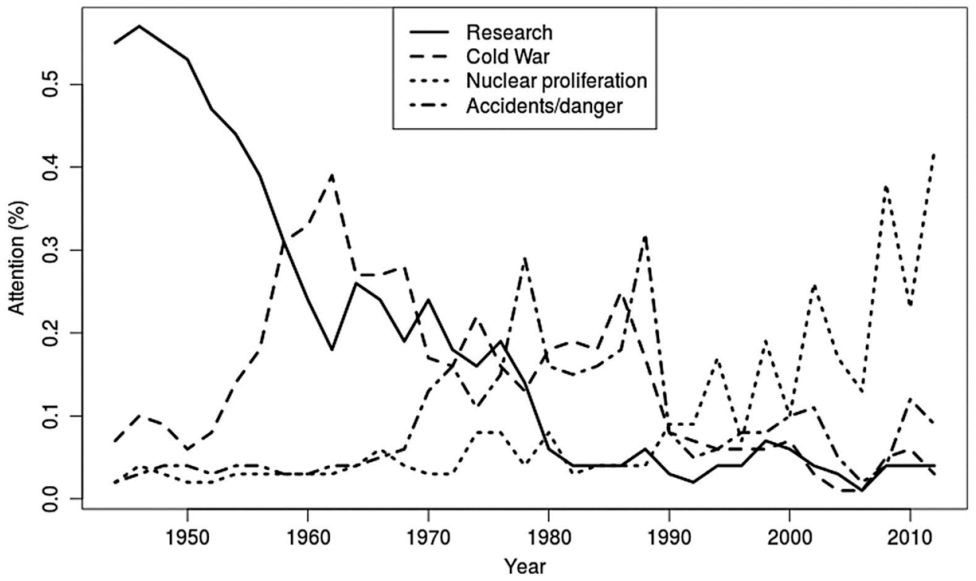
Visual communication | Human–machine communication

Kinds of data and methods

- **Text**: dictionaries, classifiers, topic models, LLMs
- **Images and video**: faces, objects, visual framing
- **Networks**: who follows, replies to or shares from whom
- **Digital traces**: logged use, web tracking, data donations
- **Simulation**: agent-based models of how opinions and information spread, now also with LLM agents
- **Research materials**: stimuli generated with LLMs; mock-up apps and social media environments, vibe-coded for experiments
- **Prediction and validation**: models that forecast behaviour, and tests of whether a measure captures what it should

Example 1 · Seventy years of nuclear news

- 51,528 *New York Times* stories on nuclear technology, 1945–2014
- Far too many for human coding
- Jacobi et al. automated the procedure via an unsupervised method called LDA-topic modelling
- The topic model finds recurring clusters of words and tracks them over time
- Research → Cold War → accidents → proliferation
- Allows to visualize attention to these topics in news over time

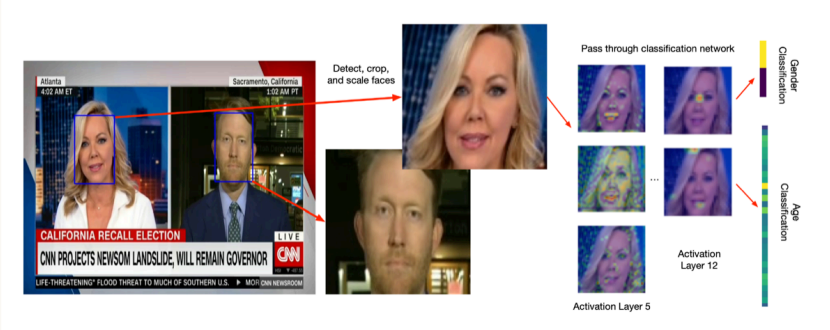


Attention to four topics over time. Jacobi, van Atteveldt & Welbers (2016).

Jacobi, C., van Atteveldt, W., & Welbers, K. (2016). Quantitative analysis of large amounts of journalistic texts using topic modelling. *Digital Journalism, 4*(1), 89–106. https://doi.org/10.1080/21670811.2015.1093271

Example 2 · Who is on German television?

- Six channels, 72 sample days, every frame between 08:00 and 24:00
- **16 million faces**: nobody can watch this much television, but a validated classifier can
- A face detector finds and crops every face; a second network estimates gender and age
- The classifier is checked against hand-coded faces first; without that step the numbers mean nothing

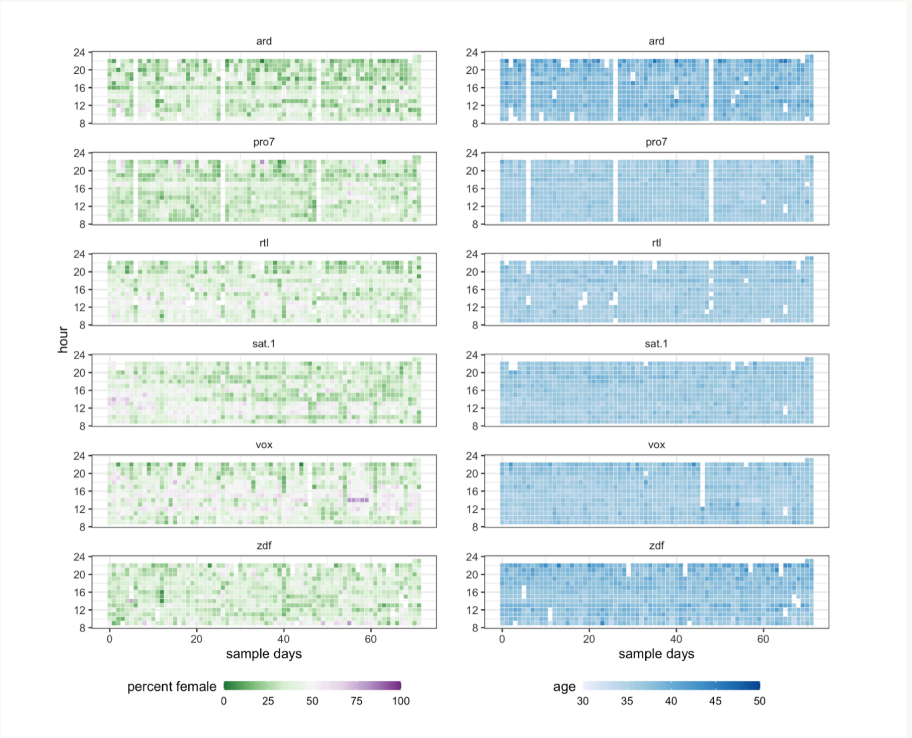


Detect, crop, classify, validate: the same four steps we will usefor text, just with pixels (images) as input.

Jürgens, P., Meltzer, C. E., & Scharkow, M. (2022). Age and gender representation on German TV: A longitudinal computational analysis. *Computational Communication Research, 4*(1), 173–207.

Example 2 · Results

- **6.3 million** of the 16 million faces were female, a proportion of **.39** (95% CI .37 to .42)
- The imbalance holds across news, sport and advertising, hour by hour and day by day
- It is also stable over time, so the pattern is structural
- Women on screen are also, on average, younger than men



Each tile is one hour of one day. Left: share of female faces. Right: estimated age.

Jürgens, P., Meltzer, C. E., & Scharkow, M. (2022). Age and gender representation on German TV: A longitudinal computational analysis. *Computational Communication Research, 4*(1), 173–207.

II · Text as data

A new kind of data

- A lot of communication is encoded in texts
- But text does not look like data we can easily analyse

SURVEY DATA

```
# A tibble: 5 × 4
     id condition sns_use well_being
  <int> <chr>       <dbl>      <dbl>
1     1 A             2.1        3.4
2     2 B             2.5        2.4
3     3 A             3.3        3.3
4     4 B             4.6        1.9
5     5 A             1.8        4.4
```

TEXT DATA

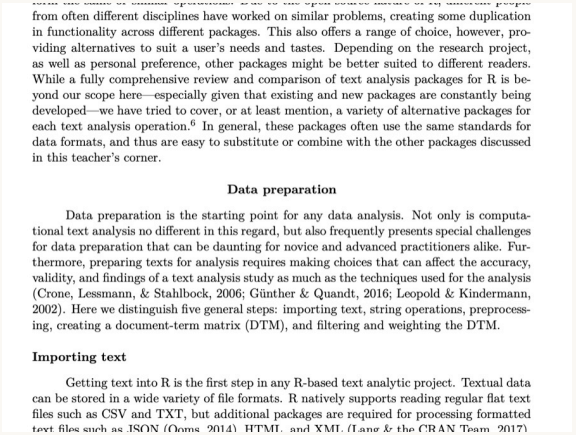
```
# A tibble: 5 × 1
  text
  <chr>
1 North Korea launched a ballistic missile into the Sea of Japan, South Korea's…
2 Tributes poured in for Colin Powell after the announcement of his death at 84.
3 An Indian couple arrived for their wedding by sailing through flooded streets…
4 Billionaire Peter Thiel is facing opposition over his plans to build a lodge …
5 Unique events that led to civilisation mean its demise could end meaning in t…
```

> Most communication is encoded in text, and text does not look like data we – or a computer – know how to analyse.

What is a text?



Hieroglyphs carved into a temple wall



from often different disciplines have worked on similar problems, creating some duplication in functionality across different packages. This also offers a range of choice, however, providing alternatives to suit a user's needs and tastes. Depending on the research project, as well as personal preference, other packages might be better suited to different readers. While a fully comprehensive review and comparison of text analysis packages for R is beyond our scope here—especially given that existing and new packages are constantly being developed—we have tried to cover, or at least mention, a variety of alternative packages for each text analysis operation.[6] In general, these packages often use the same standards for data formats, and thus are easy to substitute or combine with the other packages discussed in this teacher's corner.

Data preparation

Data preparation is the starting point for any data analysis. Not only is computational text analysis no different in this regard, but also frequently presents special challenges for data preparation that can be daunting for novice and advanced practitioners alike. Furthermore, preparing texts for analysis requires making choices that can affect the accuracy, validity, and findings of a text analysis study as much as the techniques used for the analysis (Crone, Lessmann, & Stahlbock, 2006; Günther & Quandt, 2016; Leopold & Kindermann, 2002). Here we distinguish five general steps: importing text, string operations, preprocessing, creating a document-term matrix (DTM), and filtering and weighting the DTM.

Importing text

Getting text into R is the first step in any R-based text analytic project. Textual data can be stored in a wide variety of file formats. R natively supports reading regular flat text files such as CSV and TXT, but additional packages are required for processing formatted

A printed journal article



```
    <ul class="nav navbar-nav navbar-right">
      <li>
  <a href="content.html">Content &amp; Learning Goals</a>
</li>
<li>
  <a href="overview.html">Overview</a>
</li>
<li>
  <a href="material.html">Slides &amp; Material</a>
</li>
<li>
  <a href="contact.html">Contact</a>
</li>
    </ul>
  </div><!--/.nav-collapse -->
</div><!--/.container -->
</div><!--/.navbar -->

<div id="header">


<h1 class="title toc-ignore">Computational Analysis of Digital
Communication</h1>

</div>


<p>This page contains all information and materials for the course
"Computational Analysis of Digital Communication (CADC)" developed at
the Vrije Universiteit Amsterdam. In the course, students will learn
about common computational methods in Communication Science. They will
learn how to use the <a href="https://www.r-project.org/">statistical
programming environment R</a> to:</p>
<ul>
<li>import various data formats as well as gather data from online
sources,</li>
<li>transform and wrangle data to get it ready for analysis,</li>
<li>perform text classification and text analysis (including dictionary,
classic machine learning approaches, and using large language models),
and</li>
<li>perform advanced statistical analysis.</li>
</ul>
<p><br></p>
<p><img
src="https://socialsciences.nature.com/cdn-cgi/image/quality=90/https://images.zapnito.com/uploads/d39ae7f9b96dbb3ee350deac50
<p><em><a
href="https://socialsciences.nature.com/posts/54262-computational-social-science-heralds-the-age-of-interdisciplinary-science"
source: Sutherland, 2019</a></em></p>
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```

The HTML behind a web page

- All three are **symbols in a sequence**, meaningless on their own
- Meaning arrives only when someone interprets them, in context
- Automated text analysis tries to bridge that gap

Krippendorff, K. (2018). *Content analysis: An introduction to its methodology* (4th ed.). SAGE.

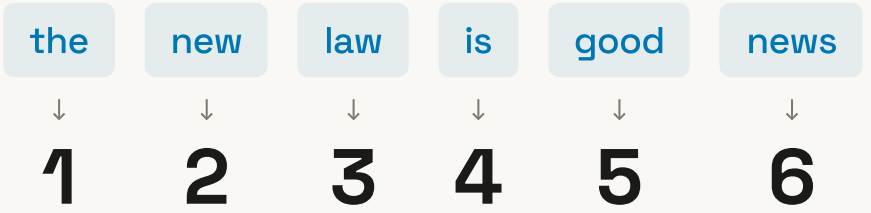
Content analysis, by hand

1. Choose the texts
2. Define the units and the categories
3. Write a codebook: rules another coder could follow
4. Code, which is where the time goes
5. Check agreement between coders, revise the rules, code again
6. Analyse frequencies, relationships, change over time

> **Step 4 hides more than it seems. The coder decodes symbols into words, words into language, and language into meaning, in a context the text does not spell out. That skill is learned: the coder has to know how to read.**

Computers don’t read text, they read numbers

- The simplest way: integer encoding
- Numbering words in order of first appearance
- Problem: Doesn’t encode much at all

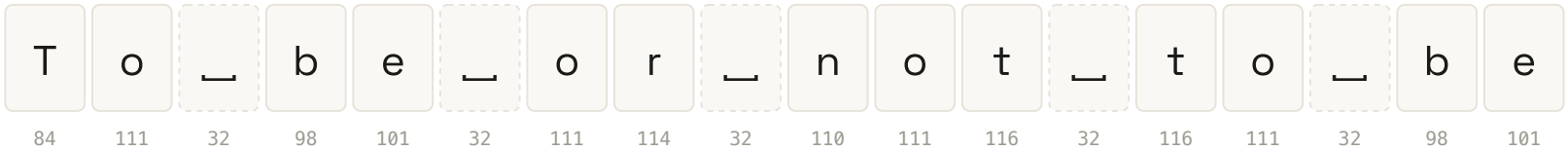


> So how do we actually turn text into numbers?

From text to numbers

01 characters | 02 tokens | 03 counts | 04 embeddings

TEXT 1 To be or not to be



18 characters → 18 numbers

A text is symbols.

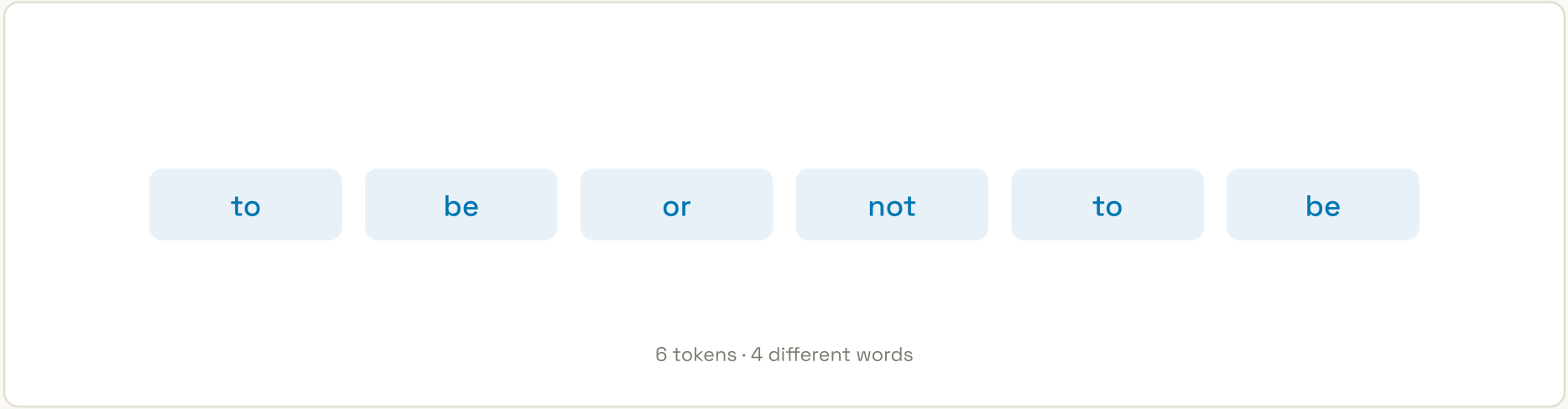
PLAYGROUND Tokeniser sandbox

1 / 5

From text to numbers

01 characters | 02 tokens | 03 counts | 04 embeddings

TEXT 1 To be or not to be



PLAYGROUND Tokeniser sandbox

Cut it into tokens.

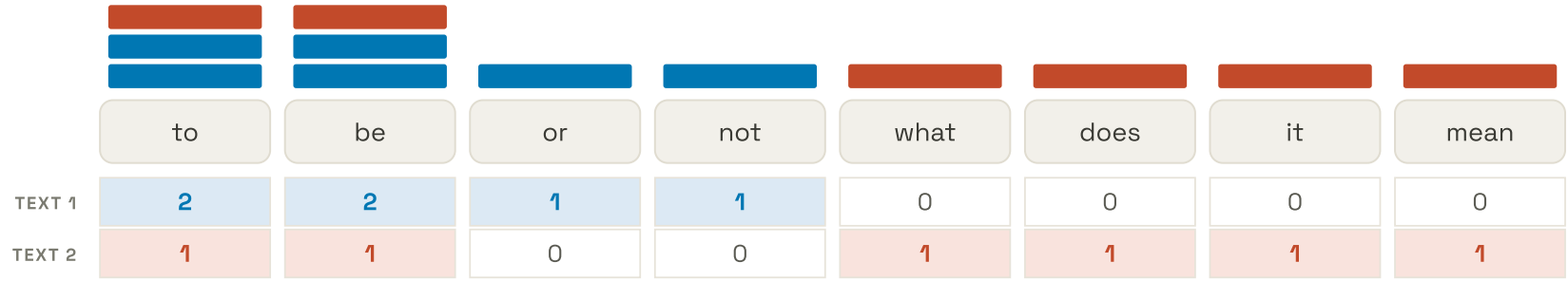
2 / 5

From text to numbers

01 characters | 02 tokens | 03 **counts** | 04 embeddings

TEXT 1 To be or not to be

TEXT 2 What does it mean to be



2 texts × 8 words — a document–term matrix

Count the words.

PLAYGROUND Tokeniser sandbox

3 / 5

From text to numbers

01 characters | 02 tokens | 03 counts | 04 embeddings

TEXT 1 To be or not to be

TEXT 2 What does it mean to be

	to	be	or	not
dim 1	−.12	.34	.21	−.63
dim 2	.48	−.19	.07	.55
dim 3	−.77	.62	−.15	.09

300 rows, and not one of them has a name

words used alike land near each other



two of those 300 rows, plotted

Or place the words.

PLAYGROUND Tokeniser sandbox

4 / 5

From text to numbers

01 characters — 02 tokens — 03 counts — 04 embeddings

TEXT 1 To be or not to be

TEXT 2 What does it mean to be

WEEKS 1–3

Counts

one row per text, one column per word

2	2	1	1	0	0
1	1	0	0	1	1
0	3	1	0	2	0

WEEKS 3–4

Embeddings

one column per word, 300 rows nobody named

.21	−.48	.77	.05
−.34	.19	.62	−.41
.07	.55	−.15	.66

Playground · the tokeniser lets you build the left one yourself

PLAYGROUND Tokeniser sandbox

Two shapes carry the whole course.

A small example text corpus

```r
texts <- tibble(
  doc  = c("A", "B", "C"),
  text = c("The new law is good news.",
           "The new law is not good news.",
           "A disaster, this law."))
texts
```

```
# A tibble: 3 × 2
  doc   text
  <chr> <chr>
1 A     The new law is good news.
2 B     The new law is not good news.
3 C     A disaster, this law.
```

> Note that these sentences say different things, but share in part the same words!

From tokens to counts

```
1  library(tidytext)
2
3  texts |>
4    unnest_tokens(word, text) |>     # one row per token
5    count(doc, word)                 # one row per document and word
```

- Three tiny texts, so every number can be checked by eye
- `unnest_tokens` splits, lowercases and drops punctuation in one call; `count` does the arithmetic
- One row per document and word is the **tidy** shape: every later step in this course is a join or a summary on it

OUTPUT

```
# A tibble: 17 × 3
  doc   word      n
  <chr> <chr> <int>
1 A     good      1
2 A     is        1
3 A     law       1
4 A     new       1
5 A     news      1
6 A     the       1
7 B     good      1
8 B     is        1
9 B     law       1
# ℹ 8 more rows
```

The document–term matrix

```
texts |>
  unnest_tokens(word, text) |>
  count(doc, word) |>
  pivot_wider(names_from = word,
              values_from = n,
              values_fill = 0)  # a word absent from a text counts 0
```

OUTPUT

```
# A tibble: 3 × 11
  doc     the   new   law    is  good  news   not     a disaster  this
  <chr> <int> <int> <int> <int> <int> <int> <int> <int>    <int> <int>
1 A         1     1     1     1     1     1     0     0        0     0
2 B         1     1     1     1     1     1     1     0        0     0
3 C         0     0     1     0     0     0     0     1        1     1
```

> The DTM is the representation we use in weeks 2 and 3.

- The same counts, turned on their side: one row per **document**, one column per **term**, a count in every cell
- Compare it with the survey table on “A new kind of data”: rows are cases, columns are variables. Text has become ordinary data
- Most cells are zero, and with real corpora almost all of them are. The matrix is **sparse**

What the matrix throws away

TEXT	LAW	GOOD	NEWS	NOT	DISASTER
A · The new law is good news.	1	1	1	0	0
B · The new law is not good news.	1	1	1	1	0
C · A disaster, this law.	1	0	0	0	1

- Five of the ten columns from the previous slide; texts A and B say opposite things and differ in one cell
- The matrix keeps which words occur and how often; it drops the order they came in. This is the **bag-of-words** assumption
- Cheap and surprisingly effective for many questions; blind to negation, irony, and anything that lives between words
- Treating *not* as a negative word does not repair it: each column is counted on its own, so *“not a disaster”* falls just as far as *“not good”*

Where the numbers come from

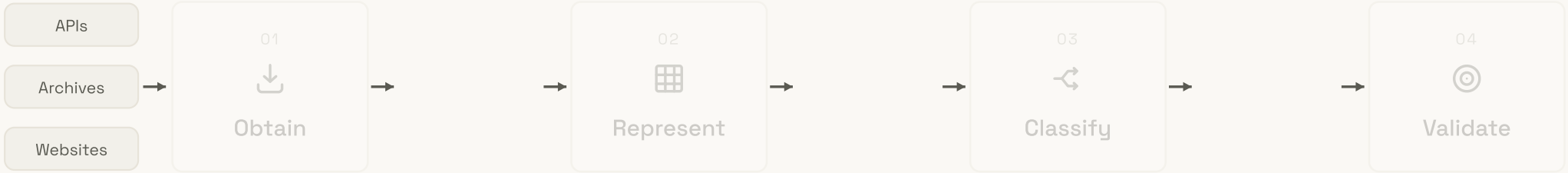
LEVEL OF LANGUAGE	IT ORGANISES	THIS COURSE USES IT IN
Morphology	how words are formed	tokens, stems, n-grams (weeks 1 to 3)
Syntax	how sentences are built	word order, mostly thrown away until week 3
Semantics	what words and sentences mean	word embeddings (week 3)
Pragmatics	how language is used in context	language models (week 4)

> **Every method in this course is a decision about which of these levels to keep.**

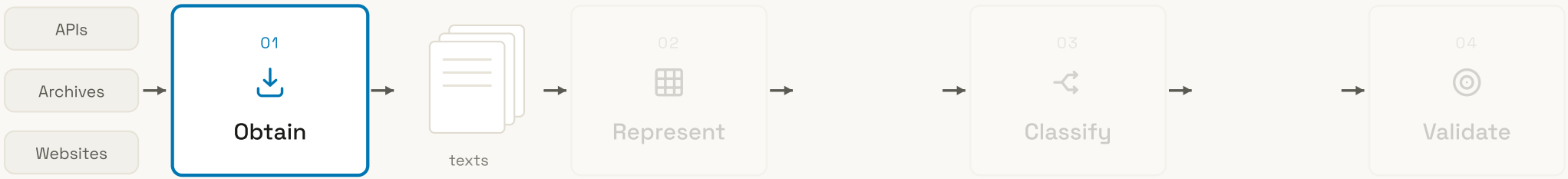
Hvitfeldt, E., & Silge, J. (2021). *Supervised machine learning for text analysis in R*. CRC Press. Chapter 1.

III · The text classification pipeline

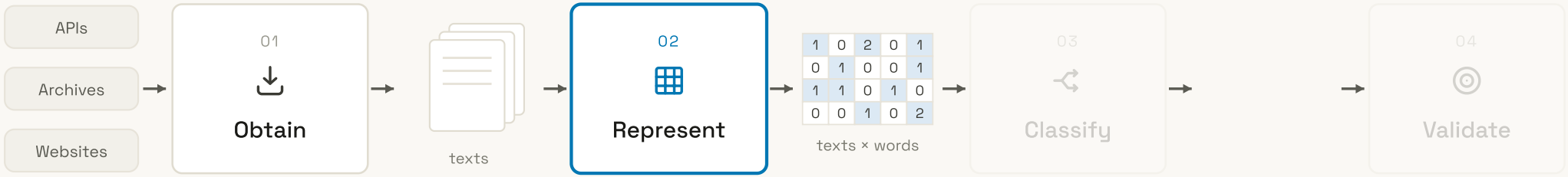
The pipeline



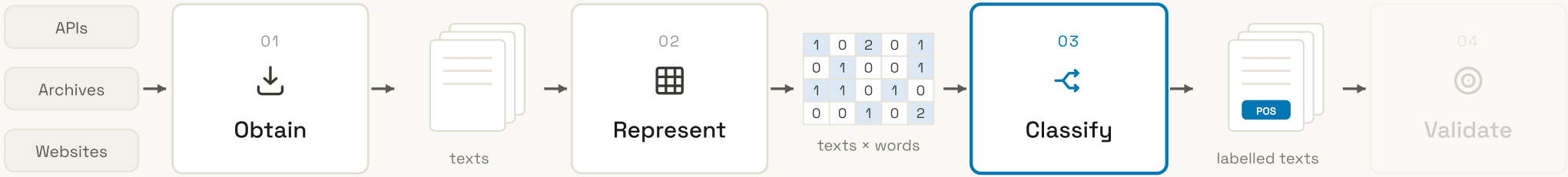
The pipeline



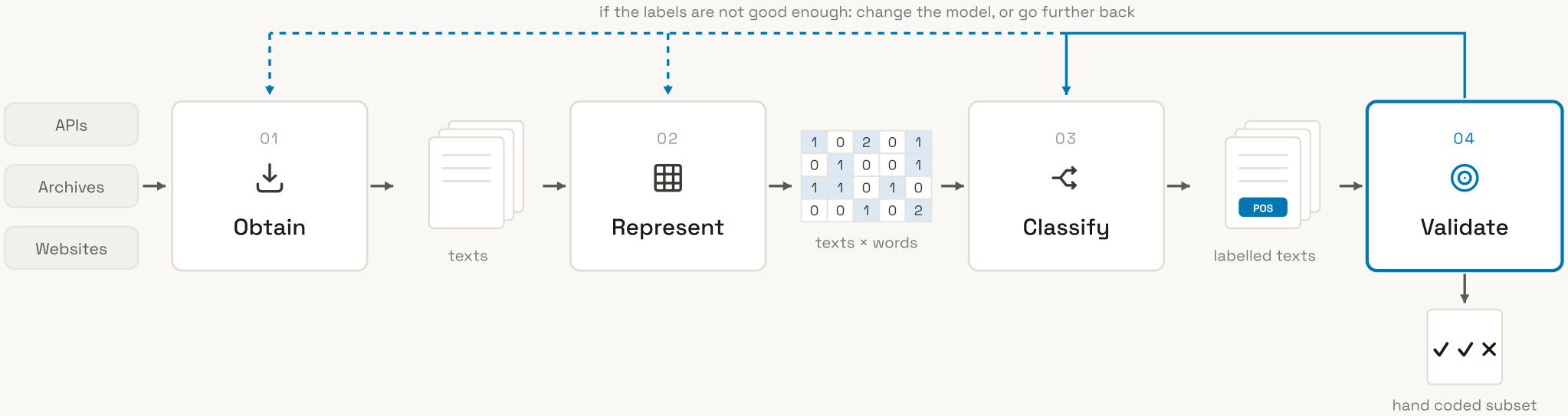
The pipeline



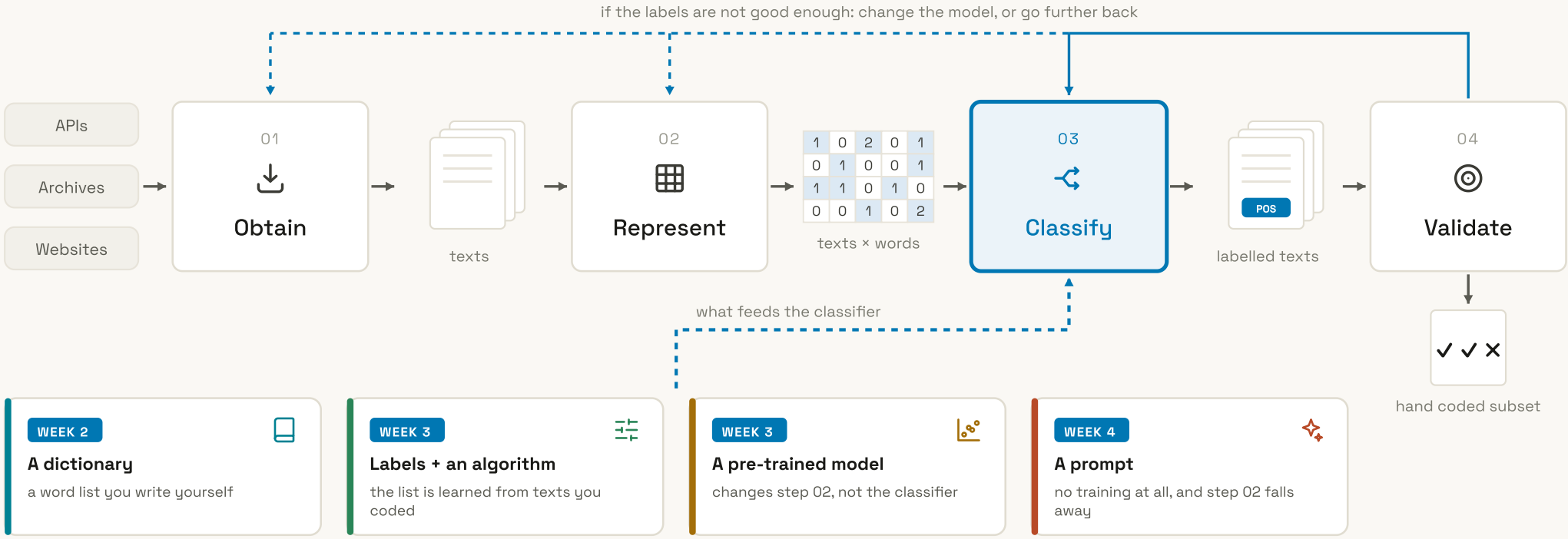
The pipeline



The pipeline



The pipeline



01 · Obtain

- **Public data sets**: parliamentary records, news corpora, research archives; ideal when they exist
- **Scraping**: websites and archives you collect yourself; trivial or very hard depending on the site, and always a legal question
- **Third parties**: APIs, commercial archives, data donations; custom formats, rate limits, terms of use that change

STEP 01

From page to text

what a scraper receives, and the part you keep

```
<div class="story">
  <p class="lede">North Korea launched a ballistic missile into
the Sea of Japan, South Korea's military said.</p>
  <div class="ad">Advertisement</div>
  <a href="/share">Share</a>
  <script src="tracker.js"></script>
</div>
```

↓

North Korea launched a ballistic missile into the Sea of Japan, South Korea's military said.

> **Obtaining text is rarely the interesting part and often the slowest. Plan for it.**

02 · Represent

- **Create features**: break text into units (words, n-grams, sentences)
- **Transform**: lowercase, strip punctuation, remove stopwords, stem
- **Select**: drop very rare and very frequent terms
- **Build the data**: the document–term matrix from Part II for counting and classic learning; dense vectors for neural models

WEEKS 1–3

Counts

one row per text, one column per word

2	2	1	1	0	0
1	1	0	0	1	1
0	3	1	0	2	0

WEEKS 3–4

Embeddings

one column per word, 300 rows nobody named

.21	−.48	.77	.05
−.34	.19	.62	−.41
.07	.55	−.15	.66

> **Every item on this list is a decision, and every decision changes the measurement. Modern models leave you fewer, but some remain.**

03 · Classify

WEEK 2

Rule-based

"If word X occurs, the text means Y."

MEANING COMES FROM

the researcher, in advance

a sentiment dictionary

NOT IN THIS COURSE

Unsupervised

"These words cluster; I decide what the cluster means."

MEANING COMES FROM

the researcher, afterwards

topic models

WEEK 3

Supervised

"Texts like these were labelled Y, so this one is probably Y too."

MEANING COMES FROM

human-coded training data

Naive Bayes, SVM, neural nets

WEEK 4

Pre-trained models

"Given everything I have read, this text is Y."

MEANING COMES FROM

the model's training corpus

GPT, Llama, Claude

> The four approaches differ mainly in where the meaning of a label comes from. Otherwise, the all try to achieve the same thing: Label a text!

04 · Validate

- Every classifier is a black box to some degree, dictionaries included, and manual coding too
- The machine does not understand the text; it predicts a label from features, and it is wrong some of the time
- So the analysis must be shown to be valid: compare its labels with a known standard
- The reference is usually a so-called **gold standard**: a subset coded by trained humans, with agreement between them checked

WEEK 2

Confusion matrix

the word list against the gold standard, texts A to C

		WORD LIST SAYS positive	WORD LIST SAYS negative
GOLD STANDARD	positive	1 text A	0
GOLD STANDARD	negative	1 text B	1 text C

Accuracy: 2 of 3 texts on the diagonal, 0.67

> **Important: A automated result only becomes a good measurement if it has been validated!**

The gold standard, next to the text

TEXT	GOLD STANDARD	WORD LIST	AGREE
A · The new law is good news.	positive	positive	yes
B · The new law is not good news.	negative	positive	no
C · A disaster, this law.	negative	negative	yes

> Example: If we just count positive words (as simple method) to deduce whether or not the text is positive, we get Text B wrong. Scaled to many text, comparing to a gold standards helps to find these errors.

The computational mind

- In this course, we want to teach you a “computational mindset”
- Humans do not easily make the leap from a social science question to something a computer can automate
- But thinking computationally is a key skill today and in the future, and this course hopefully teaches it

WHAT WE WANT TO KNOW

Did Dutch news cover the housing crisis negatively?

A question about meaning. It has no fixed answer yet.

WHAT A COMPUTER CAN DO

Give every article one label from a fixed set, the same way each time.

A procedure with an input, an output and a check.

The same question, framed twice

NOT YET A PROBLEM

- “Is the housing coverage negative?”
- Negative to whom, measured how, compared with what?

A PROBLEM YOU CAN RUN

- Input: one article
- Output: negative / neutral / positive
- Criterion: agrees with a trained human coder
- Check: on 100 articles nobody used to build it

The pipeline runs on problems, not on questions

Did **Dutch news coverage** of housing get **more negative between 2015 and 2025**?

1 / 6

The pipeline runs on problems, not on questions

Did **Dutch news coverage** of housing get **more negative between 2015 and 2025**?

The pipeline runs on problems, not on questions

Did **Dutch news coverage** of housing get **more negative between 2015 and 2025**?

YOUR DECISION

Which texts?

every article mentioning woningmarkt, four outlets

THE MACHINE'S JOB

What counts as negative?

one label per article, defined in a codebook

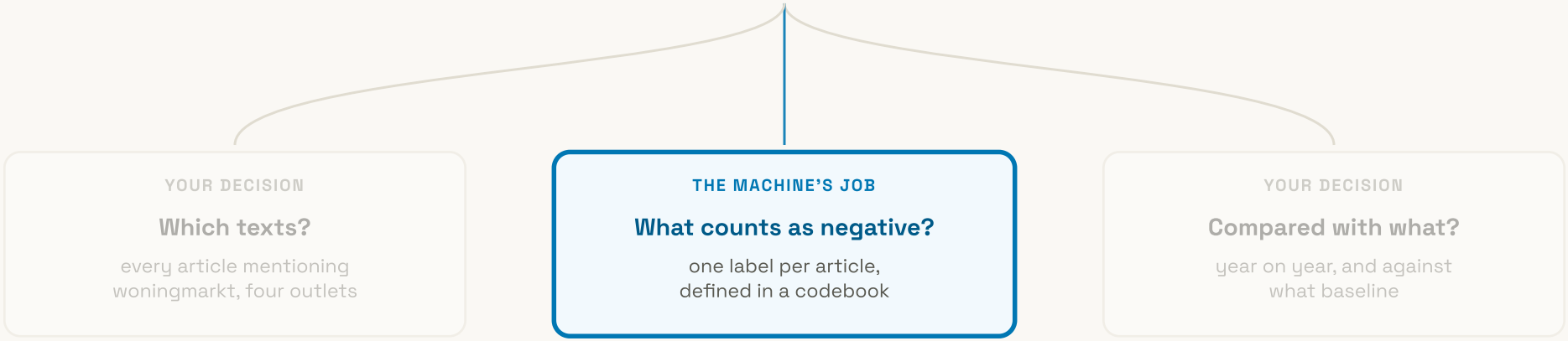
YOUR DECISION

Compared with what?

year on year, and against what baseline

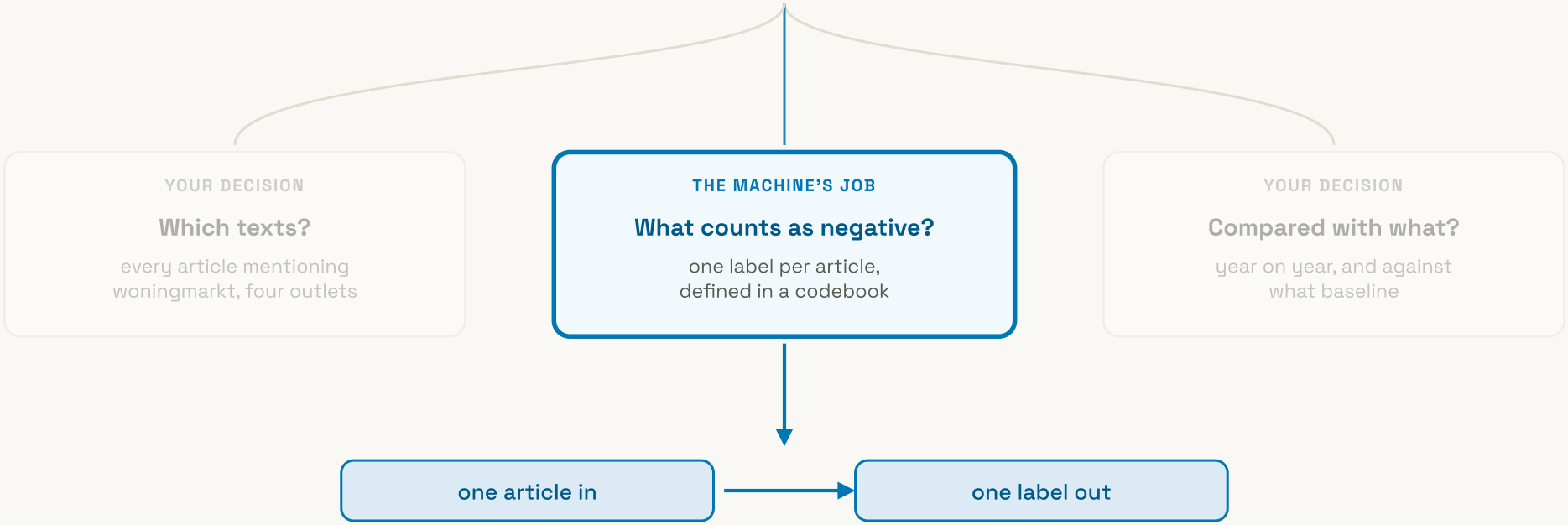
The pipeline runs on problems, not on questions

Did **Dutch news coverage** of housing get **more negative between 2015 and 2025**?



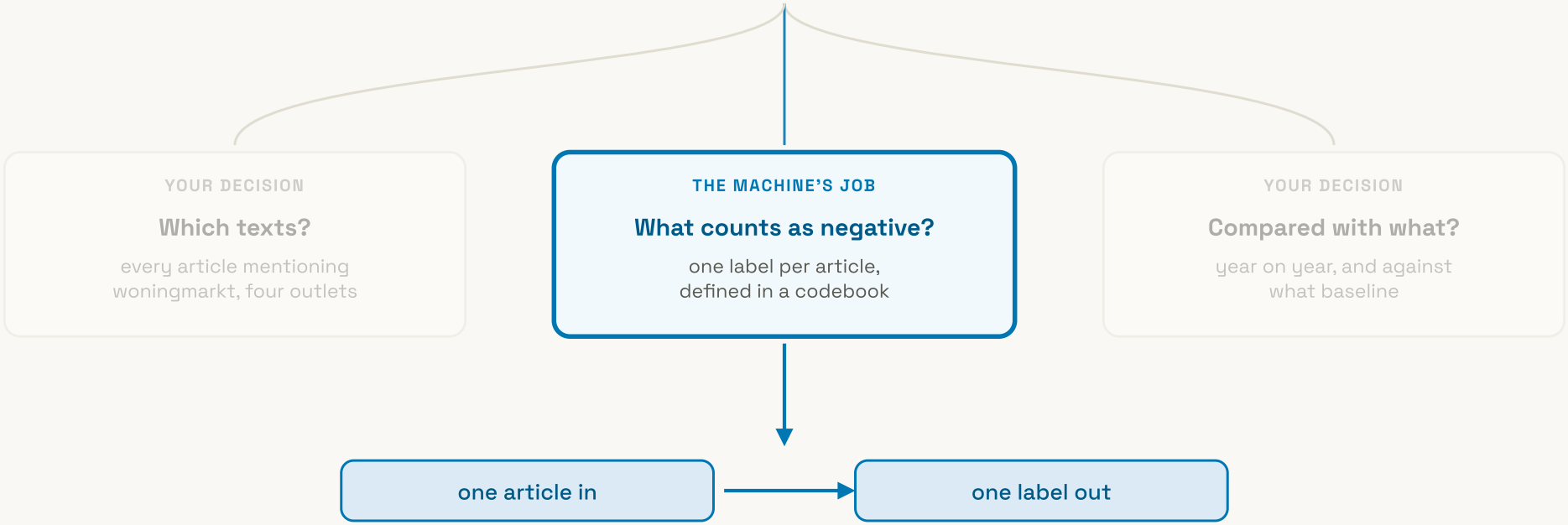
The pipeline runs on problems, not on questions

Did **Dutch news coverage** of housing get **more negative between 2015 and 2025**?



The pipeline runs on problems, not on questions

Did **Dutch news coverage** of housing get **more negative between 2015 and 2025**?



A computational problem, stated properly

1. **Input**: what exactly goes in? One article, one sentence, one post? Whose?
2. **Output**: what exactly comes out? A label, a score, a number per month?
3. **Success criterion**: what would count as good enough, decided before you run anything
4. **Failure check**: how would you find out that you were wrong?

> **Note: In each lab, you will answer these questions, on paper, before you start to code a text analysis procedure.**

IV · A short history of text analysis

Timeline of natural language processing

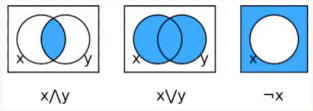
Bayes' theorem

- Published two years after Thomas Bayes's death, edited by Richard Price
- How to update a probability given evidence — the basis of **Naive Bayes** classifiers

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Boolean logic

- George Boole's "Laws of Thought" turns reasoning into algebra: AND, OR, NOT
- The basis of digital circuits, programming languages and keyword search



1763 — 1843 — 1854 — 1936

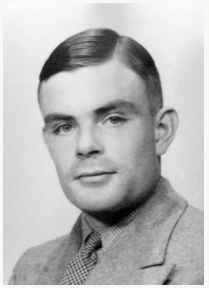
The first algorithm

- Ada Lovelace's "Note G": a program for Babbage's Analytical Engine
- She foresees machines that operate on symbols of any kind, not only numbers



The universal machine

- Alan Turing's abstract machine: a limitless tape, a head that reads and writes one symbol at a time
- Anything computable can be computed by it — the blueprint of the computer

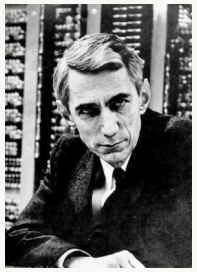


Timeline of natural language processing

1948

Information theory

- Claude Shannon: information measured in bits, uncertainty as entropy
- English modelled as letter and word probabilities (**n-grams**) — the first statistical language model
- Anthropic's Claude is named after him



1949

Weaver's memorandum

- Warren Weaver frames machine translation as code-breaking and statistics
- 1954: the Georgetown–IBM demo translates 60 Russian sentences; MT is declared three to five years away



1950

Can machines think?

- Turing's "Computing Machinery and Intelligence" proposes the imitation game — the **Turing test**
- A "child machine" that learns from experience instead of being programmed

1957

Grammar theories

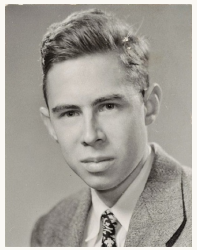
- Noam Chomsky's "Syntactic Structures": a finite set of rules generates infinitely many sentences
- Three decades of symbolic NLP follow: hand-written grammars and parsers



Timeline of natural language processing

1958 — The perceptron

- Frank Rosenblatt's first trainable **neural network**: weighted inputs, a threshold, a learning rule
- Minsky & Papert (1969) show what one layer cannot learn; neural nets go out of fashion



1966 — The first dictionary: the General Inquirer

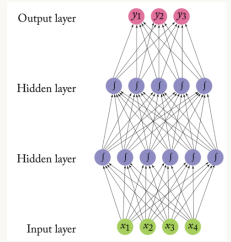
- Stone, Dunphy, Smith & Ogilvie (Harvard): word lists per category, applied to texts by a mainframe — the first computer **content analysis**, and the **dictionary approach** of week 2
- Same year: Weizenbaum's ELIZA imitates a therapist by pattern matching; the ALPAC report ends US funding for machine translation

1970s–80s — Knowledge-based NLP

- SHRDLU (1970) follows commands in a blocks world; expert systems
- Meaning is encoded by hand in ontologies: concepts, properties, relations
- Works in toy worlds, does not scale to real text

1986 — Backpropagation

- Rumelhart, Hinton & Williams: multi-layer networks can be trained
- Neural nets return — but data and compute are still too scarce for language

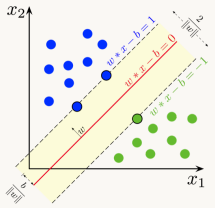


Timeline of natural language processing

1990s

The statistical turn

- IBM (Brown et al., 1990): translation learned from parallel text, not from rules
- Hidden Markov models, Naive Bayes spam filters, SVMs (1995), LSTMs (1997)
- Corpora replace grammars; evaluation on held-out data becomes the norm



2001–2003

LIWC & topic models

- **LIWC** (Pennebaker, 2001): the 1966 dictionary idea goes mainstream in psychology — word lists that count emotional and cognitive categories
- **LDA** (Blei et al., 2003): topics as distributions over words
- Bengio et al. (2003): the first neural language model

2012

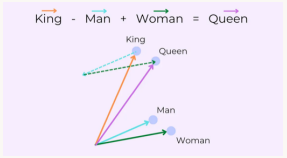
Deep learning takes off

- AlexNet wins ImageNet by a wide margin, trained on GPUs
- Learned features beat hand-crafted ones; big data and GPUs make the decade possible

2013

word2vec

- Mikolov et al. learn **word embeddings** efficiently from raw text
- Vector arithmetic captures meaning: king − man + woman ≈ queen
- GloVe (2014) follows; pre-trained embeddings lift every downstream task

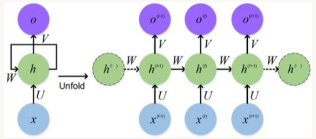


Timeline of natural language processing

2014–2016

Sequence models

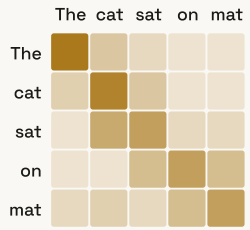
- Sequence-to-sequence learning with LSTMs: an encoder compresses the input, a decoder generates the output (Sutskever et al., 2014)
- Bahdanau et al. (2014) add **attention**: the decoder looks back at every input word



2017

Attention & Transformers

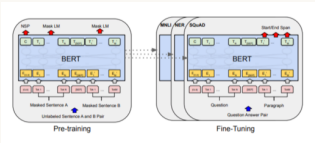
- Vaswani et al., “Attention Is All You Need”: self-attention only, no recurrence
- Trains in parallel on huge corpora — the architecture behind every model since then, including ChatGPT



2018

BERT & GPT

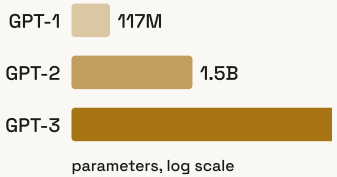
- GPT (OpenAI) predicts the next word; BERT (Google) fills in masked words
- Pre-train once on the web, fine-tune for any task: **transfer learning** becomes the recipe



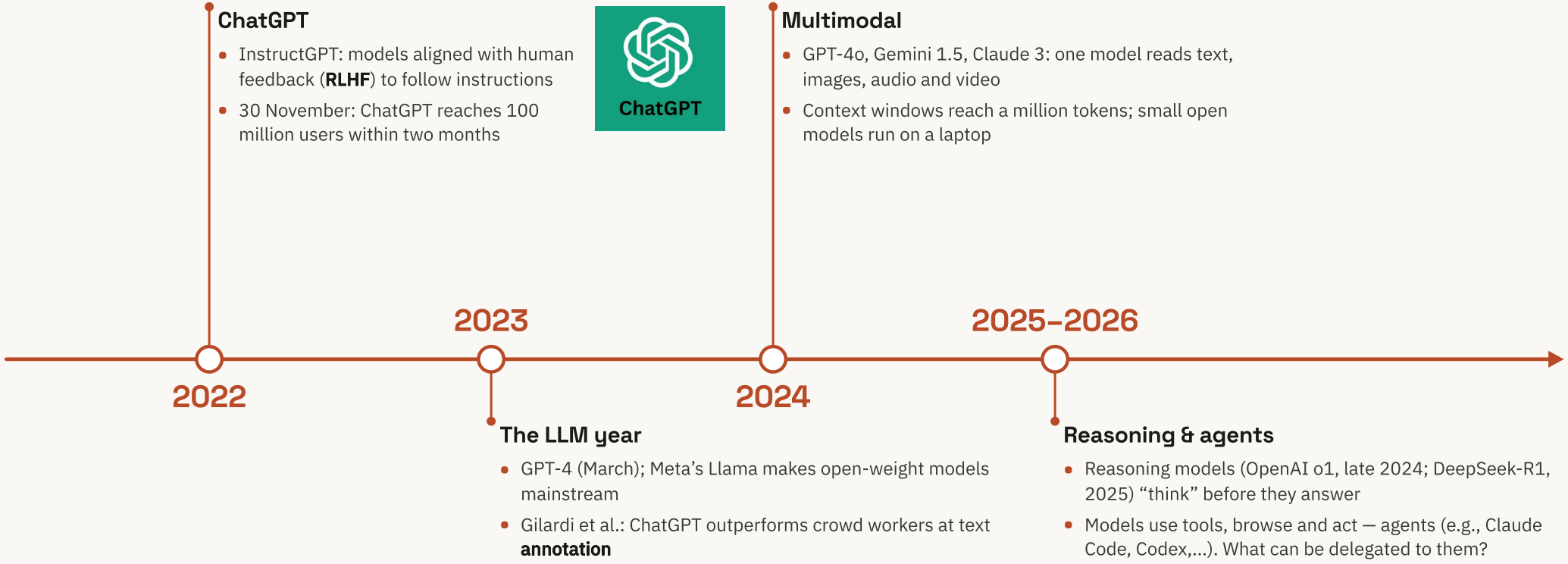
2019–2020

Scaling

- GPT-2 (1.5B parameters) writes coherent paragraphs; GPT-3 (175B) learns new tasks from a few examples in the prompt
- **In-context learning**: no fine-tuning, just instructions



Timeline of natural language processing



This course: Same task, four generations

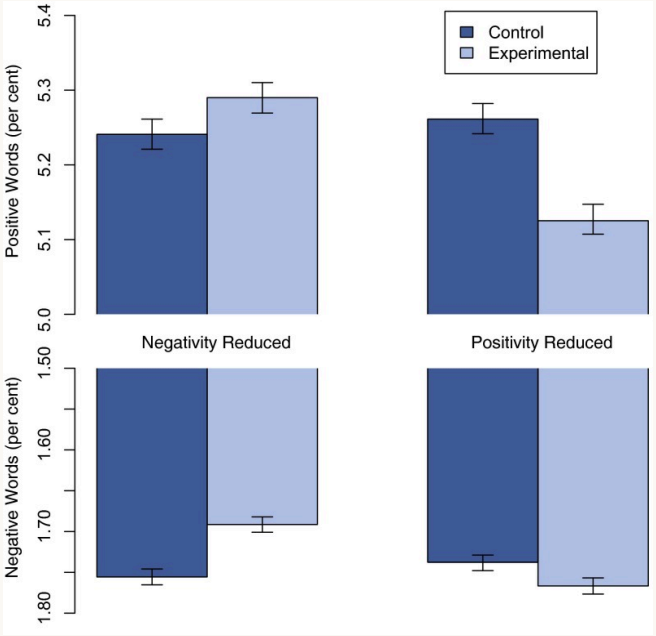
- In the next four weeks, we will go through some milestones of this timeline
- The same measurement task runs through weeks 1 to 4
- We will hand-code a gold standard in week 2
- Four methods scored against it: counting, a dictionary, a trained classifier, a language model

> **So by week 4 you can say which method is better, for the task, on this data, rather than which is newer.**

V · Ethics in text analysis

A controversial study (Kramer et al., 2014)

- 2012: Facebook alters the news feeds of **689,003 users** for one week: fewer positive posts for some, fewer negative for others
- A dictionary (LIWC) decides which posts count as positive or negative: one emotion word from its lists is enough
- The same dictionary measures the users' own posts: their emotion words shift in the same direction; the effect is real and tiny
- The users were not asked, informed or debriefed
- Published in *PNAS* in 2014; the reaction was immediate



> **The terms of service made it legal; whether it was right is a separate question.**

Kramer, A. D. I., Guillory, J. E., & Hancock, J. T. (2014). Experimental evidence of massive-scale emotional contagion through social networks. *PNAS, 111*(24), 8788–8790. https://doi.org/10.1073/pnas.1320040111

What went wrong in the Facebook study

ETHICS

- **No consent**: the data policy users had accepted did not mention research until four months after the study
- **No debriefing**: participants were never told whether they were in it
- **Deliberate manipulation** of emotions, with no opt-out and no exclusion of vulnerable users
- **No independent review**: Cornell's board judged that its researchers had only analysed results

VALIDITY

- **A blunt dictionary**: one emotion word decides, so "not happy" counts as positive
- **Words are not feelings**: the outcome is the share of emotion words in a post, which also follows norms of what one posts
- **Overclaiming**: "massive-scale emotional contagion" from shifts of about a tenth of a percentage point

Hill, K. (2014, June 30). Facebook added "research" to user agreement 4 months after emotion manipulation study. *Forbes.*
Verma, I. M. (2014). Editorial expression of concern. *PNAS, 111*(29), 10779. https://doi.org/10.1073/pnas.1412469111

General ethical challenges in computational science

- **Privacy**: personal data, data breaches, and the question of what you send to a company's model
- **Bias**: models inherit the biases of their training data, and validation is the only way to see it
- **Transparency**: complex models are hard to explain, which makes them hard to hold to account
- **Manipulation**: the same tools shape opinion and behaviour, with or without consent
- **Cost**: energy for training and running models, and ownership of models and data

Van Atteveldt, W., & Peng, T.-Q. (2018). When communication meets computation. *Communication Methods and Measures, 12*(2–3), 81–92. https://doi.org/10.1080/19312458.2018.1458084

Four principles

- **Respect for persons**: treat people as autonomous; honour their wishes
- **Beneficence**: understand and improve the risk–benefit profile of the study
- **Justice**: risks and benefits should fall on the same people
- **Respect for law and public interest**: privacy, copyright, access rights

> **The first three come from the Belmont Report (1978), the fourth from the Menlo Report (2012). Salganik applies all four to digital research.**

Dittrich, D., & Kenneally, E. (2012). *The Menlo Report: Ethical principles guiding information and communication technology research*. U.S. Department of Homeland Security.
Salganik, M. J. (2018). *Bit by bit: Social research in the digital age*. Princeton University Press. Chapter 6.

Principles for computational communication research

PRINCIPLE	IN PRACTICE
1 · Privacy and data security	Consent where you can; know where your data go
2 · Fairness	Look for bias in data and models; validate
3 · Transparency	Prefer interpretable models; share complex ones
4 · Care with experiments	Consent and debriefing, or assess the harm
5 · Environmental impact	Test small first; most tasks need no giant model
6 · Cross-cultural sensitivity	Models do not cross languages for free
7 · Engagement	Involve the communities your research touches

Can you use AI in this course?

NO AI

Part 1 of every lab, the check-ins and the exam

Guided practice, and tests of what you can do yourself: reading output, reading code, judging a design. Code completion in RStudio, such as GitHub Copilot, is switched off too.

WHY

You can only judge a model's output with knowledge you built without one

AI PERMITTED, DISCLOSED

Part 2 of every lab, the homework and the project

Use whatever tools help you, and say how: one line in Lab A, the full disclosure in the homework and the project. We then ask for your judgement: what you kept, what you rejected, and how you checked it.

NEVER

Work you cannot defend without the output in front of you

Four kinds of AI use

KIND OF USE	WHAT IT MEANS	WHAT WE ASK OF YOU
Instrument	The model does a defined task whose output enters your work: classifying texts, writing code you run, producing labels you analyse	A methods-style disclosure: model and version, how the prompt was designed, validation against your own judgement
Assistant	Editing, grammar, translation, debugging, help searching; the content is yours	Which tool, for what, and what you kept, changed or rejected
Thinking partner	It works on your own notes and drafts, and stress-tests ideas that are already yours	What you discussed and what came of it; you can defend every claim without the output
Surrogate	The model produces the work and you put your name on it	**Not allowed**: it falls under the VU fraud and plagiarism regulation

VI · How this course works

The team

- **Assoc Prof. Dr. Philipp K. Masur**: coordinator, lecturer and lab teacher (labs in Dutch); contact through Canvas
- **Lab teachers**: Dr. Noon Abdulqadir (labs in English) and Gianni Quaedvlieg, MSc (labs in Dutch); six groups, two each, assigned on Canvas

The weekly rhythm

- **Monday · lecture**: the problem and the respective the method
- **Tuesday and Thursday · labs**: your coding in R, in six groups; mandatory
- **Sunday · homework**: started in Thursday's lab, finished at home; one per week, weeks 1 to 4
- **Slides and labs** on the course website
- **Communication and submission** via Canvas

> **Note: Each lab opens with a short check-in: not graded, but six of eight are required to submit.**

Assessment

COMPONENT	WEIGHT	WHAT IT IS
Exam	40%	Week 5, in person, without an AI to ask. Reading output and code, judging whether a measurement can be believed. All content taught in the lectures and the labs!
Homework	30%	Four assignments, one per week, each started in Lab B: your code, its output, short reflections on your own results, and an AI disclosure if you used it. All four required for a pass
Group project	30%	Weeks 6 to 8, a study of your own in a group, ending in a mini conference

> **Part 1 of every lab, the check-ins and the exam are the unaided lane. Part 2 of every lab, the homework and the project permit AI use with a proper disclosure.**

This week

- **Tuesday · Lab A**: tidy wrangling (filter, select, mutate, group, summarise) on European survey data
- **Thursday · Lab B**: tokens, counts and a document–term matrix, then State of the Union speeches corpus
- **Sunday · homework 1**: New Year's resolution tweets (clean, count, compare), started in Lab B
- **Before Thursday · Playground**: the tokeniser sandbox, two minutes
- **Reading**: van Atteveldt & Peng (2018); *R for Computational Social Science*, ch. 2–5

> Before Tuesday: work through *Before the first lab* on the course website. The labs start on your own laptop.

Setup help · one page on the course website

PART 1 · BEFORE TUESDAY · UNDER AN HOUR

1. Check your laptop: Windows or macOS; a Chromebook or an iPad cannot run RStudio
2. Install R, then RStudio
3. Change three settings in RStudio
4. Install the course packages
5. Make a course folder and run the setup check: done when no line says `[fix]`

PART 2 · FOR WHEN THINGS GO WRONG

- Downloads, zip files and file extensions
- Folders, paths, and where R looks for a file
- The keys R needs, such as `|`, `~` and `{`
- What the common error messages mean

> **Find it as Setup help at the bottom of every page of the course website, step by step for Windows and macOS. Stuck? Send a message on Canvas with the error as text and the output of the setup check.**

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Questions?

> If anything is unclear, ask now, or on Tuesday, with a keyboard in front of you.

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